

# Mathematics Extension 2

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**General  
Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your Centre Number and Student Number at the top of this page and on the Question 11 Writing Booklet attached

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**Total marks:  
100****Section I – 10 marks** (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

**Section II – 90 marks** (pages 7–14)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

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1 What are the square roots of  $-3 + 4i$ ?

A.  $1 - 2i$  and  $-1 + 2i$

B.  $1 + 2i$  and  $-1 - 2i$

C.  $3 + i$  and  $-3 - i$

D.  $-3 + i$  and  $3 - i$

## Section II

**90 marks**

**Attempt Questions 11–16**

**Allow about 2 hours and 45 minutes for this section**

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

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### Question 11 (10 marks)

- (a) The complex numbers  $w$  and  $z$  are given by  $w = 3e^{i\pi/2}$  and  $z = e^{i\pi/3}$ . **2**

Find the modulus and argument of  $wz^2$ .

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- (b) The complex number  $z$  is given by  $5 + 2i$ .

Find, in Cartesian form:

(i)  $z^2$  **1**

(ii)  $z^{-1}$  **2**

**End of Question 11**